IKT200-Assignment 3- Solution

Task 1: Rule-based systems

a) "Add rules to complete the stoplight example. Handle the case of a flashing yellow light (proceed with caution). Handle the case of a power outage (caution is required)." *Existing rules:*

- RedRule - IF red light is on THEN stop.
- GreenRule - IF green light is on THEN go.
- YellowFarRule- IF yellow light is ON AND vehicle is moving 30 miles an hour AND distance to stoplight is greater than 150 feet THEN stop.
- YellowCloseRule - IF yellow light is ON AND vehicle is moving 30 miles an hour AND distance to stoplight is less than 150 feet THEN go.
- RedFlashingRule - IF red light flashing THEN caution is required
- CautionRule - IF caution is required AND vehicle is moving THEN stop
- ProceedRule - IF caution is required AND vehicle is stopped THEN proceed with caution

New rules:

- YellowFlashRule - IF yellow light is flashing THEN proceed with caution
- PowerOutage - IF red light is off AND green light is off AND yellow light is off THEN caution is required

Other rules could be added to handle cases such as a stalled car, or an approaching ambulance.

b) "Add rules to handle a broken stoplight control mechanism. For example, what if both the green and red lights are on?"

- BrokenLight1 - IF red light is ON AND green light is ON THEN stoplight is broken
- BrokenLight2 - IF red light is ON AND yellow light is ON THEN stoplight is broken (NB: This is the case in normal operation for Norwegian traffic lights ...)
- BrokenLight3 - IF yellow light is ON AND green light is ON THEN stoplight is broken
- BrokenLight4 - IF stoplight is broken THEN caution is required

c) "Add rules to handle invalid input facts. For example, the facts red light is on, and red light is flashing are both in working memory."

- BadInput1 - IF red light is ON AND red light is flashing THEN input is invalid
- BadInput2 - IF green light is ON AND green light is flashing THEN input is invalid
- BadInput3 - IF yellow light is ON AND yellow light is flashing THEN input is invalid
- BadInput4 - IF input is invalid THEN message ("One or more inputs are invalid.") And HALT

Task 2: Turing test

What is the Turing test all about?

The Turing test is a test described by Alan Turing in 1950. The test deals with a machine's ability to display intelligent behavior similar to that of a human, or not recognizable as a machine, to a

human. Turing suggested that a human will evaluate using the natural language in the conversation between a human and a machine. The human evaluator will be aware that one of the two parties in the conversation is a machine, and participants will be separated from each other. The conversation will be limited to a text with a keyboard and a screen so that the result will not depend on the machine's ability to recognize word speech. If the evaluator cannot differentiate the machine from human (Turing originally proposed that the machine would convince man 70% of the time after five minutes of conversation), the machine has passed the test. The test does not check for the ability to give correct answers to the question, just how closely the responses would be like that of human.

Task 3: Internet Principles

Answer the following questions (Textbook Chapter 3)

a) Say a little about the IP layer, and focus on the limitations.

   Data is sent in packets marked with recipient and sender. The sender machine often does not know the way to the receiver. There is no guarantee that packages will arrive or that they will arrive in the correct order. You cannot send more than one packet at a time.

b) How can TCP improve the challenges of IP?

   Can make sure that the packages arrive (by receiving status feedback and resending failures). Can make sure data volumes greater than 1 package appears in the right order.

c) What are the differences between UDP and TCP?

   TCP is connection oriented - when a connection is established, data can be sent bidirectional. UDP is a simpler, connectionless Internet protocol.

d) Describe IPv4 and give an example of what an address might look like.

   Internet Protocol Version 4 ("IPv 4") consists of 32 bits, ie . 2 ^ 32 addresses, which is approx. 4 billion unique addresses. Difficult for people to read such large numbers, therefore the addresses are grouped into 8 bits, 4 times, rendered decimal. Example, 192.168.0.1 is a common address for home router.

e) Describe IPv6 and give an example of what an address might look like.

   Internet Protocol version 6 ("IPv 6") uses 128 bits for addressing (as opposed to ipv4 which uses 32 bits.) Grouped into 16 bits, 8 times. Each group is split again into 4 bits, 4 times displayed hexadecimal. Example, 2001: 0db8: 85a3: 0042: 0000: 8a2 e: 0370: 7334

f) What are MAC addresses?

   All devices connected to a network have Media Access Control ("MAC") addresses in addition to IP addresses. MAC addresses are globally unique, but it is not so common to use it outside the local network.

g) What are ports and how do they work in relation to IP addresses?

   Can compare it with the following: The IP address indicates the company we should send something to, and port indicates the department of the company. Each computer connected to a network has an IP address and data is transferred from one computer to another through channels, or ports, that are assigned specific numbers.

h) What are Application Layer Protocols? Give 2 examples.

It is the highest/outermost layer of communication protocols. Often used to communicate to for Internet of Things ("IoT") devices. Some common application layer protocols: - HTTP - HTTPS (encrypted HTTP) - SMTP (email)

Task 4: Beacons

a) What are beacons? Describe in your own words.

Beacons are small devices that enable accurate indoor positioning (equivalent to what GPS manages outdoor). Beacons transfer small amounts of data (unique IDs) via Bluetooth low energy (BLE) up to ca. 50 meters. Beacons are usually powered by small batteries, but they can be plugged into a USB port. In addition to dedicated beacons, mobile phones, tablets and PCs with BLE support act as beacons, with opportunity for both to send and receive beacon signals.

b) We have talked about two beacon standards: iBeacon and Eddystone. Write a few sentences about each of them.

iBeacon is the first and currently most widely used of the two communication protocols. Developed by Apple, it was originally supported by iOS and has deep integrations with the iPhone / iPad . Note that the iBeacon profile works on other operating systems as well. Eddystone is a new and open communication protocol developed by Google with Android users in mind. Much of how Eddystone works is the same as for iBeacon , but it has some extended functionality. Eddystone exists as three different types of protocols: Eddystone -UID, Edd ystone -URL, and Eddystone -TLM.

c) Give an example of how beacons can be used to solve a task.

There are many different areas to use beacons. For example:

https://twocanoes.com/10-awesome-things-you-can-do-today-with-ibeacons/